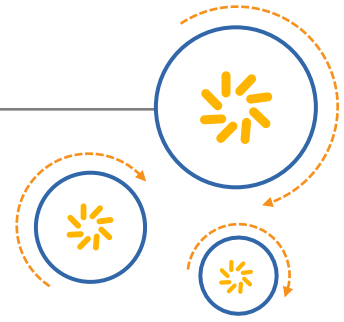




Qualcomm Technologies, Inc.



WGR7640 IC GNSS RF Receiver Design Guidelines

September 2016

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Qualcomm Technologies, Inc.
5775 Morehouse Drive
San Diego, CA 92121
U.S.A.

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Revision history

Revision	Date	Description
B	September 2016	Update to 'E' part
A	August 11, 2015	Initial release

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1 Overview

1.1 Purpose

This document provides a description of the WGR7640 IC chipset capabilities. Not all features are available, nor are all features supported in the software.

NOTE: Enabling some features may require additional licensing fees.

1.2 Acronyms, abbreviations, and terms

Table 1-1 provides definitions for the acronyms, abbreviations, and terms used in this document.

Table 1-1 Acronyms, abbreviations, and terms

Term	Definition
APQ	Application Processor Qualcomm
CSP	Control Switching Point
GLONASS	Global NAVigation Satellite System
GNSS	Global Navigation Satellite System
GPS	Globas Positioning System
LO	Local Oscillator
PLL	Phase-locked Loop
PMIC	Power Management Integrated Circuit
RF	Radio Frequency
SSBI	Signal-Signal Beating Interference
WGR	WLP GPS Receiver
WLP	Wafer-level Package

1.3 WGR7640 within the APQ chipset

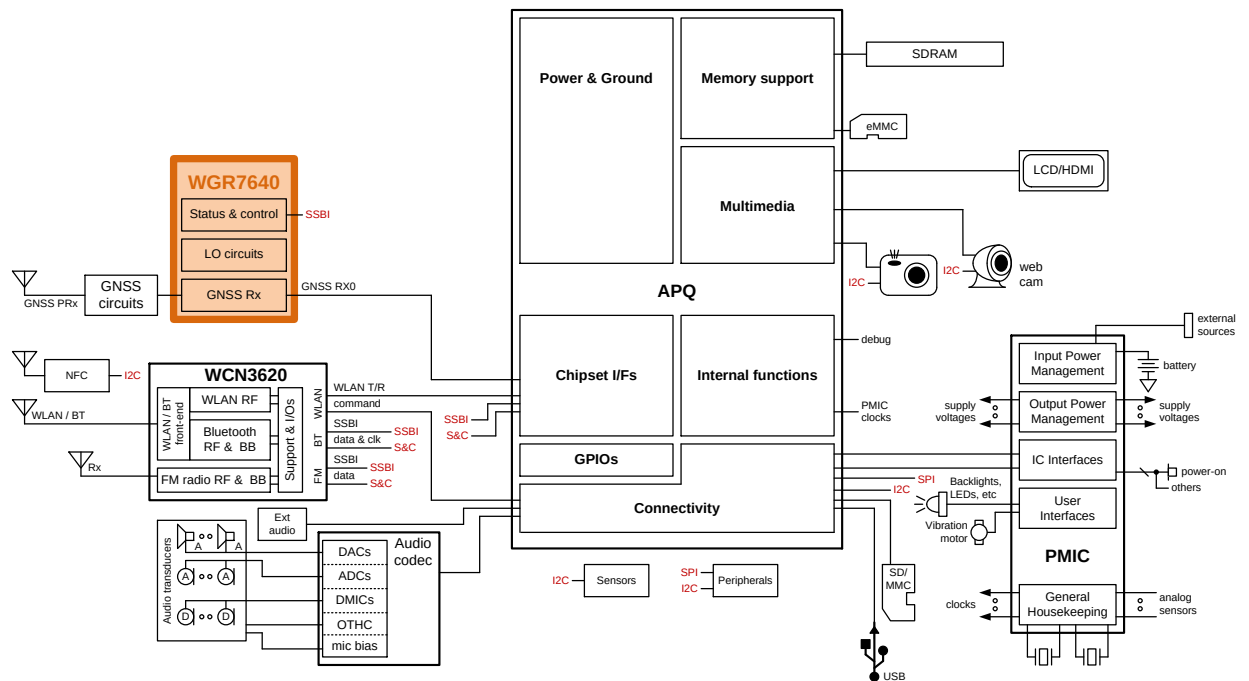


Figure 1-1 WGR7640 within the APQ chipset

1.4 Overview of WGR7640 topics

This document provides an introduction to the WGR system, followed by three major sections: RF receivers, baseband interface, and support circuits (as shown in Figure 1-2).

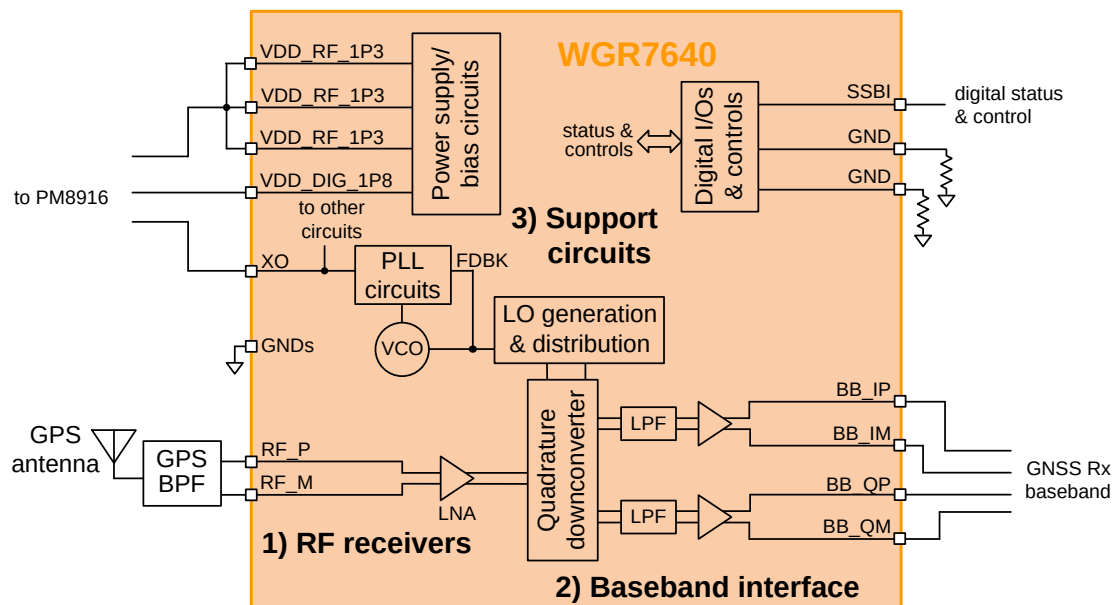


Figure 1-2 WGR7640 top-level layout guidelines

1.5 WGR7640 features

Salient features:

- Small package: 2.07 mm × 1.51 mm × 0.63 mm
- 65 nm wafer-level package (WLP)
- 17-pin CSP
- GNSS stand alone RF receiver
- Compatible with Callisto (Gen8A) GPS digital baseband engine
- Fully integrated synthesizer PLL, VCO, and loop filter components
- Power reduction features for low-power consumption
- Single-line serial bus interface (SSBI)
- The chipset solution includes the APQ + WGR7640 ICs

2 WGR7640 System

2.1 GNSS receiver system block diagram

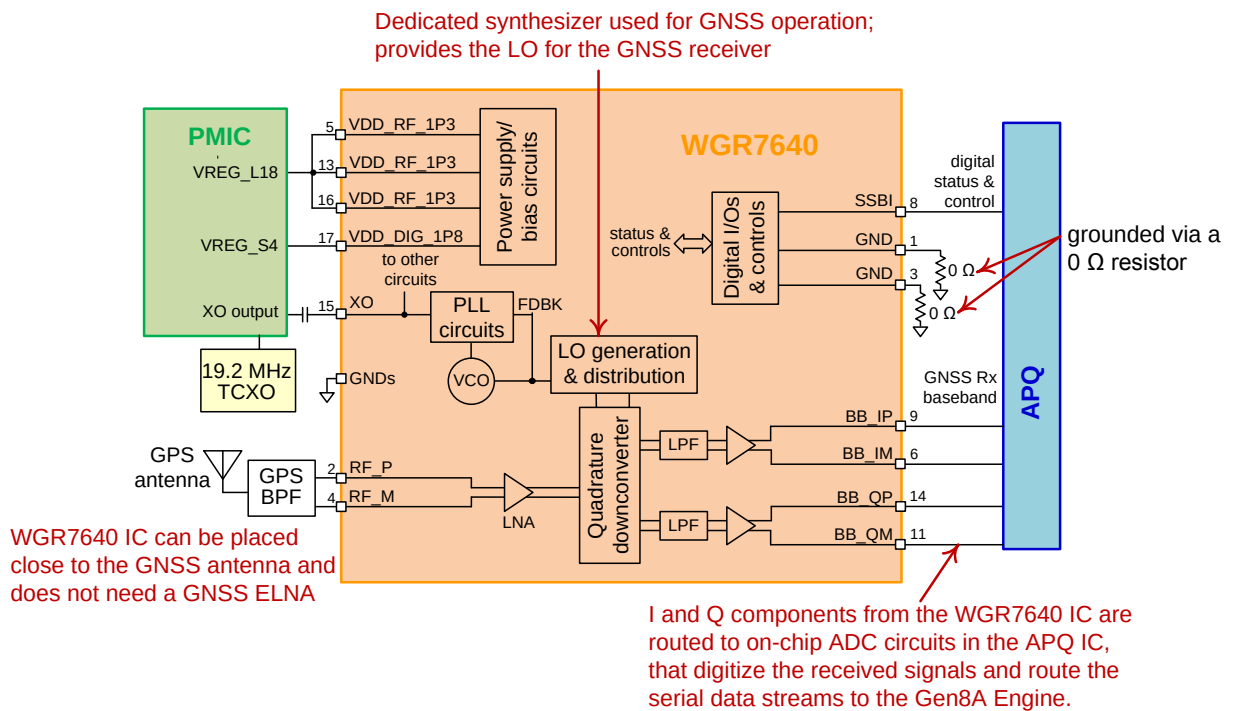


Figure 2-1 GNSS receiver system block diagram

3 WGR IC Details – RF

3.1 WGR RF receivers high-level comments

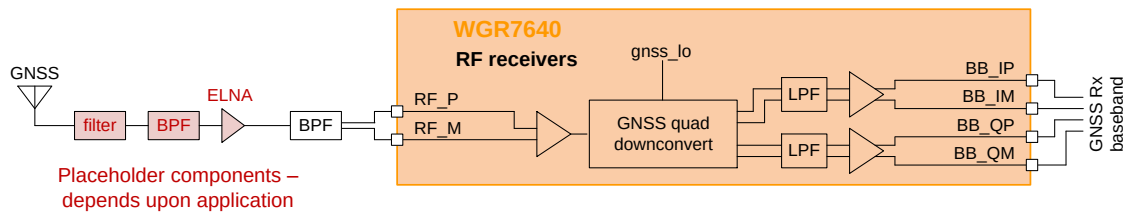


Figure 3-1 WGR RF receivers

- Receiver is implemented in 65 nm RFCMOS process which accommodates high-frequency, high-precision analog circuits and low-power CMOS functions
- Independent receive path from RF to baseband (I/Q)
- Optimized for standalone operation
- Consists of differential RF inputs supporting primary GPS, GLONASS, Compass, Galileo, and QZSS navigation satellite systems. Supported frequency bands are shown in Table 3-1.

Table 3-1 Supported frequency bands

Satellite system	Frequency (MHz)
GPS L1/QZSS	1575.42 ± 1 MHz
GPS L1C, Galileo E1, Compass B1-BOC	1575.42 ± 2 MHz
GPS L1 wide	1575.42 ± 10 MHz
GLONASS R1	1602 ± 4 MHz
Compass B1	1561 ± 2 MHz

3.2 WGR7640 RF receiver pin assignments

GNSS RF ports are grouped together (purple squares)

- Sensitive RF signals
- RF front-end (filters, etc.) located in this direction as seen in the top view

GNSS baseband (green squares)

- Sensitive analog signals
- Opposite side away from RF pins

- APQ source located in this direction as seen in the top view

1 GND		2 RF_P		3 GND
	4 RF_M		5 VDD_ RF_1P3	
6 BB_I_M		7 GND		8 SSBI
	9 BB_I_P		10 GND	
11 BB_Q_M		12 GND		13 VDD_ RF_1P3
	14 BB_Q_P			
15 XO		16 VDD_ RF_1P3		17 VDD_ DIG_1P8

Figure 3-2 WGR7640 RF receiver pin assignments

4 WGR IC Details – Baseband Interfaces

4.1 WGR Rx baseband interfaces with Application Processor Qualcomm (APQ) IC

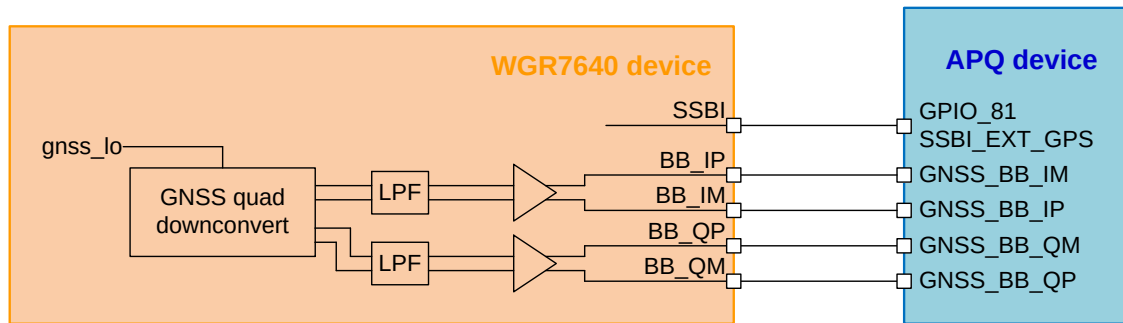


Figure 4-1 WGR Rx baseband interfaces with APQ IC

The baseband interface has two differential pairs: in-phase (I) and quadrature (Q):

- The I and Q baseband outputs are sensitive analog signals.
- Route the I and Q signals as phase-critical differential pairs.
- The resistance and capacitance on each pair should be equal.
- Avoid crossing these traces; if necessary, cross them only near the APQ end points.
 - ❑ Isolate the device from digital logic and clock traces with ground all around; treat similarly to controlled-impedance traces.
 - ❑ The length of the I/Q traces between the APQ and WGR should be equal and should not exceed 8 inches.

4.2 WGR RF GNSS input ports – schematic

Optional:

- An external LNA (not required)
- Space and pads for an external LNA preceded by a second bandpass filter (to allow addition of these components if necessary)

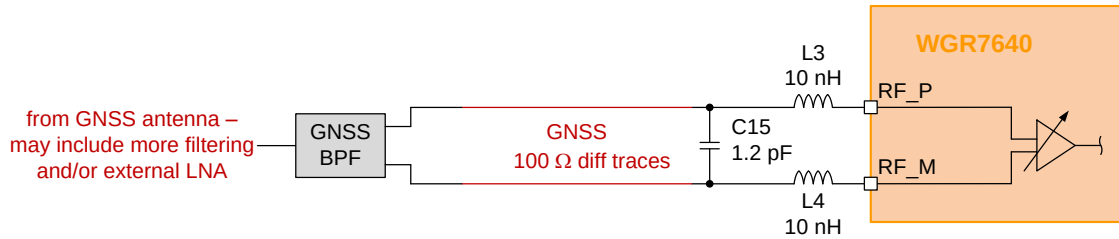


Figure 4-2 WGR RF GNSS input ports

4.3 WGR RF GNSS input ports – layout guidelines

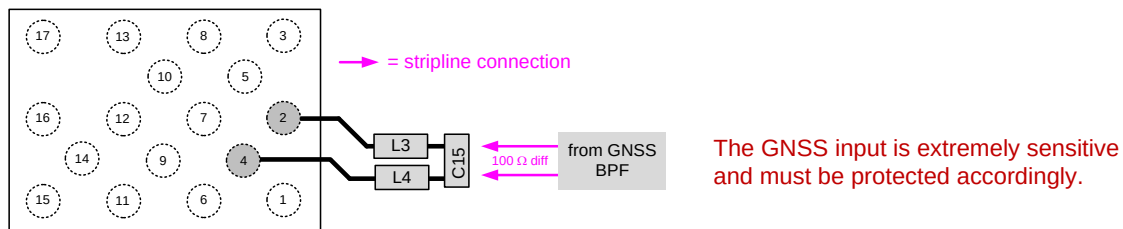


Figure 4-3 WGR RF GNSS input ports

- Traces from the filters (pink arrows) must use controlled impedance techniques – microstrip or stripline – designed to provide 100 Ω differential (nominal). Stripline provides higher isolation since it is surrounded by ground planes.
- The components are close to the WGR pins; routing is short and direct.
- If needed, the areas directly below all component pads and signal traces are cleared of metal on layer 2 to minimize parasitic capacitance (not shown).
- Inductors should be placed in ways that limit mutual coupling.
- The complementary paths within each differential pair should be as symmetrical as possible to preserve their phase balance and maintain common-mode rejection.
- Where possible, isolate matching components and traces from all other circuits and traces using coplanar ground fill (not shown).
- Trace capacitances between WGR input pins and matching components should be kept very low.
- Some extra components are included in initial layouts to increase matching flexibility but might prove unnecessary after careful testing and evaluation.
- Specific RF grounding guidelines should be followed:
 - ❑ The ground side of the RF components must connect as directly as possible to the nearest ground reference and should connect to multiple ground fills on multiple layers to provide the best grounding.
 - ❑ Do not allow long, thin traces to connect RF components to ground – the added trace inductance could disrupt circuit performance.
 - ❑ Microvias from the ground pads directly to the PCB ground plane are recommended.
 - ❑ A total ground inductance of less than 100 pH is desired.

4.4 GNSS receiver matching procedure

- Measure the differential $100\ \Omega$ input return loss (Sdd11) looking into the WGR device's matching network.
- Iteratively adjust the matching network with minor component changes to achieve the desired performance, making trade-offs between the following objectives as needed:
 - ❑ Center the passband and provide sufficient bandwidth.
 - ❑ Minimize the nominal noise figure using sensitivity measurements.
 - ❑ Standard values of inductors and capacitors can be selected.
 - ❑ The type and size of inductors can affect noise figure.

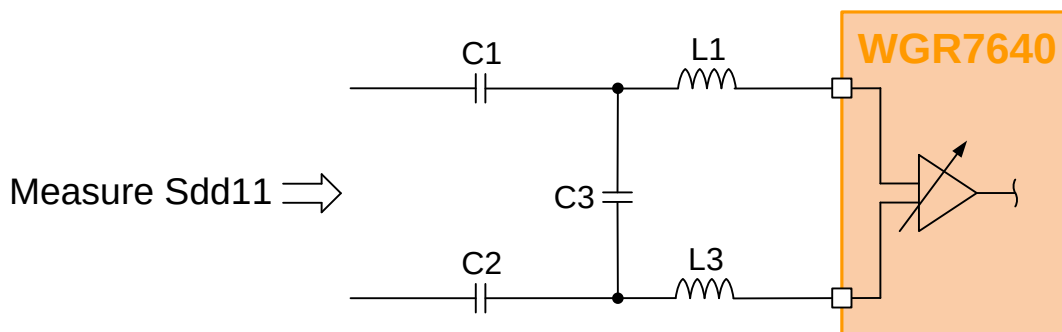


Figure 4-4 GNSS receiver matching

5 WGR IC Details – Support Circuits

5.1 WGR support circuits high-level comments

- WGR pins and circuits that provide secondary support functions to the RF transceiver include:
 - ❑ GNSS LO synthesizer and related
 - ❑ 19.2 MHz XO input
 - ❑ Digital status and control signals
 - ❑ Single-wire serial bus interface (SSBI)
- Grounds
 - ❑ Pins 1 and 3 – grounded via a 0 Ω resistor

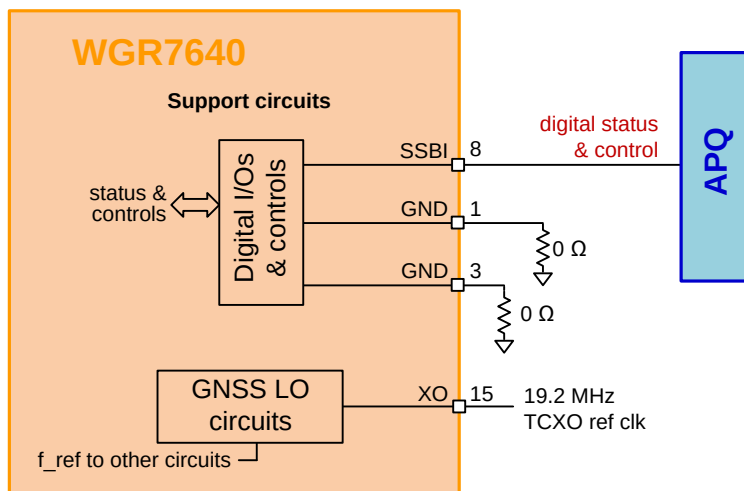


Figure 5-1 WGR support circuits

5.2 GNSS LO synthesizer

- The integrated local oscillator (LO) generation and distribution circuits are driven by internal VCOs to support various modes to yield highly flexible quadrature LO outputs to the GNSS down converter.
- The WGR7640 IC has a dedicated synthesizer used for GNSS operation. This synthesizer provides the LO for the GNSS receiver.

- For the WGR7640 IC, an external 19.2 MHz input signal obtained from the PMIC device is required to provide the synthesizer frequency reference to which the PLL is phase- and frequency-locked.
- The WGR7640 IC integrates all of the PLL loop filter components on-chip. With the integrated PLL synthesizers, the WGR7640 IC has the advantage of more flexible loop bandwidth control, fast lock time, and low integrated phase error.

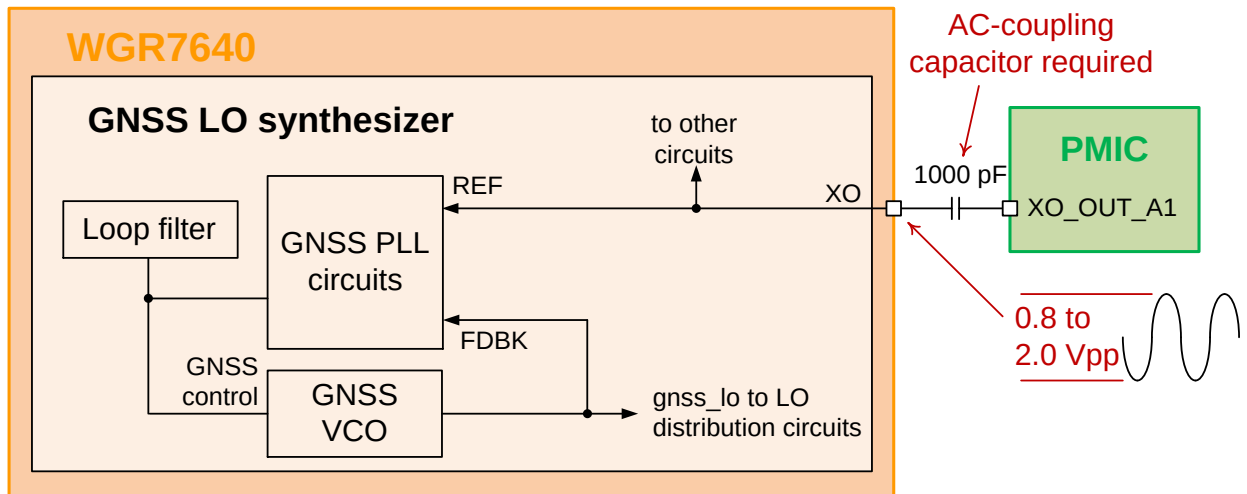


Figure 5-2 GNSS LO synthesizer

5.3 WGR digital status and control

- Power (VDD_DIG_1P3) – Use the same supply that is used for APQ digital I/Os to ensure logic compatibility and prevent latchups.
- SSBI allows efficient initialization, GNSS mode and parameter controls, and programming verification:
 - ❑ The APQ device's SSBI controller is the master while the WGR circuit is the slave.
 - ❑ SSBI is clocked at 19.2 MHz (reference clock frequency), so good layout techniques are extremely important.
- The SSBI signal is decoded and used to set on-chip control signals to the GNSS functional blocks. GNSS status is also reported to the APQ device through these circuits and SSBI.

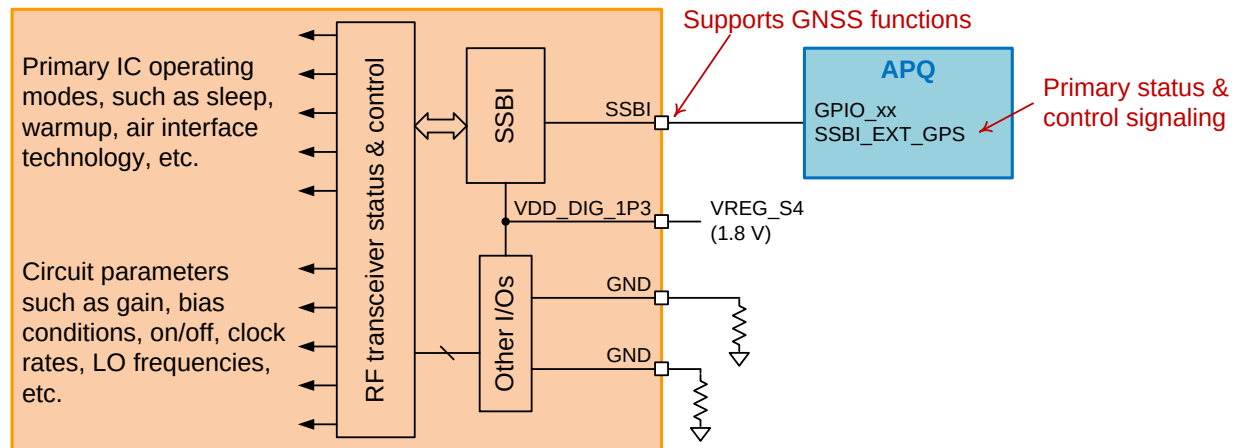


Figure 5-3 WGR digital status and control

6 Top-level Design Topics

6.1 Top-level parts placement WGR

1. Place the WGR section as close as possible to the GNSS antenna.
2. Digital devices and traces must not be placed near the top edge because they generate noise that could couple onto antennas, degrading performance (self-jamming/spurs).
3. Isolate the power management circuitry – especially SMPS circuits – from the WGR circuits.
4. The 19.2 MHz clock trace between the WGR7640 and PM8921 should be well isolated.
5. Allow space for shields. Specific shielding guidelines are given next.

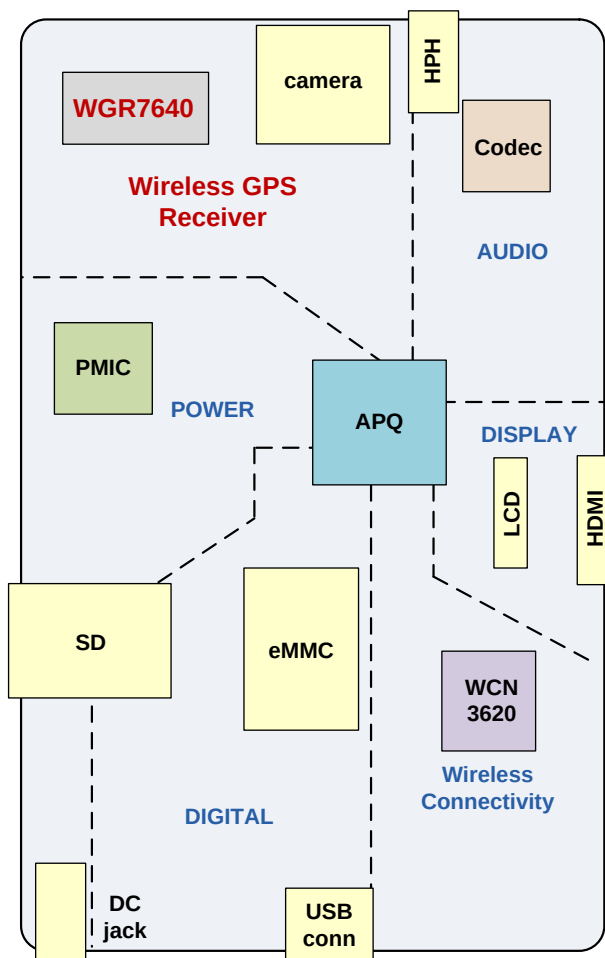


Figure 6-1 Top-level parts placement WGR

6.2 Shielding recommendations

- Shield the WGR7640 IC, its matching components, RF front end components, digital (including the APQ device), and PMIC circuits.
- Metal cans are better than metallized plastic.
- Recommended shield partitioning:
 - Preserve Tx-to-Rx isolation
 - WGR IC and its Rx matching components in one shield and RF front end components in another shield
 - Do not locate WGR matching inductors too close to shield walls (this might cause EM coupling and inductor de-Q)

6.3 WGR DC power topology

- The number of bypass capacitors required in a final design depends upon the PCB layout:
 - Initial designs should begin with “extra” series components (such as resistors, inductors, or beads) and bypass capacitors.
- After extensive test and evaluation, their numbers might be reduced prior to production.
- Ground bypass capacitors with vias that are connected directly to the internal PCB RF ground plane.
- Avoid thin, high-inductance traces.

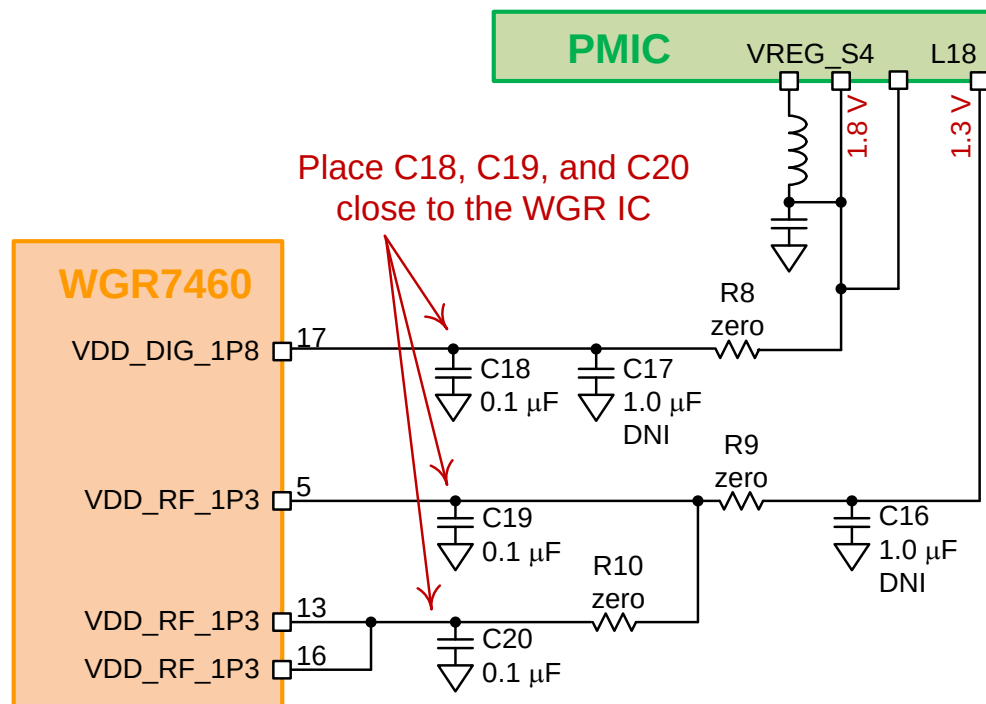
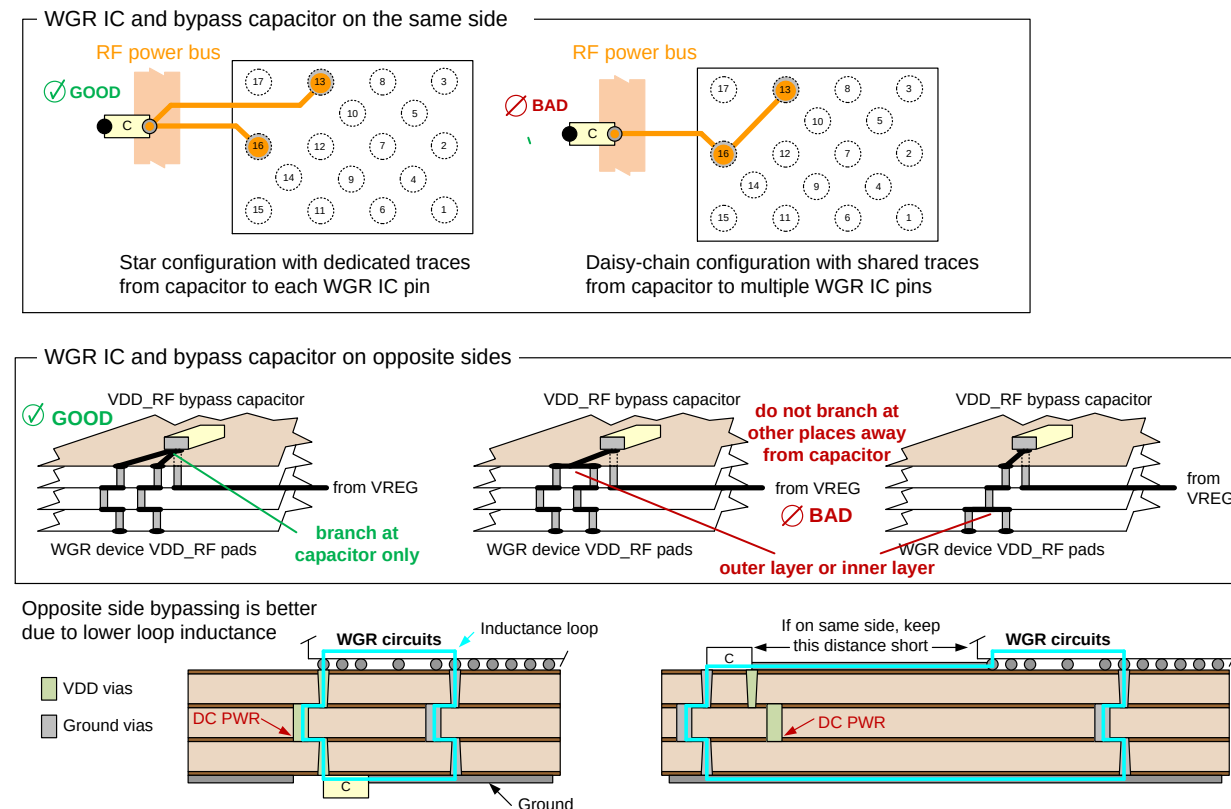


Figure 6-2 WGR DC power topology

6.4 WGR IC DC routing and bypassing – general



6.5 WGR ground connections

Proper grounding is crucial:

- Dedicate at least one layer for ground – layer 5 is shown for example
- Ground plane – common point referenced by all handset circuits
- Fill unused space on all layers to provide robust layer-to-layer ground
- Connect bypass caps directly to surface ground fill, with multiple vias to layer 5 ground

Keep WGR ground pins separate from all other circuit grounds until they converge on layer 5:

- Via each set as directly as possible to the layer 5 ground plane.
- Include as much ground fill as possible to each layer between 1 and 5 below the WGR device, connecting each stack of vias to the WGR ground fill areas.
- Use a large copper mass to provide the best possible electrical ground and thermal conductivity – both needed for WGR performance.

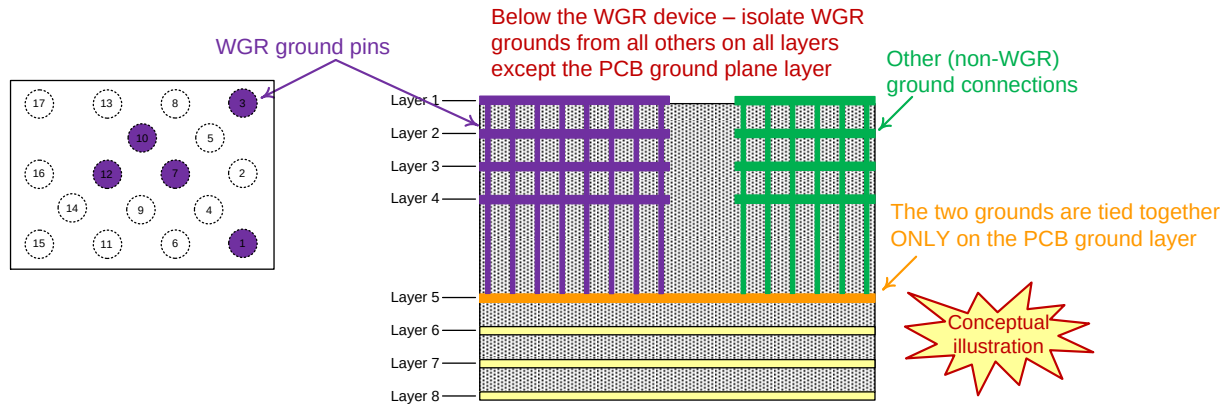
**Figure 6-3 WGR ground connections**

EXHIBIT 1

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